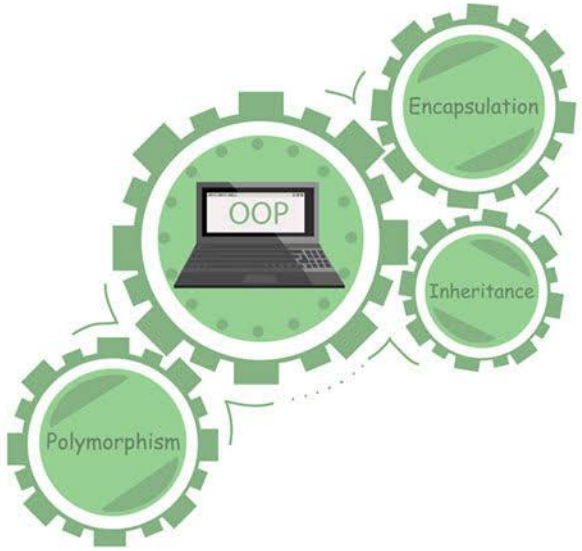
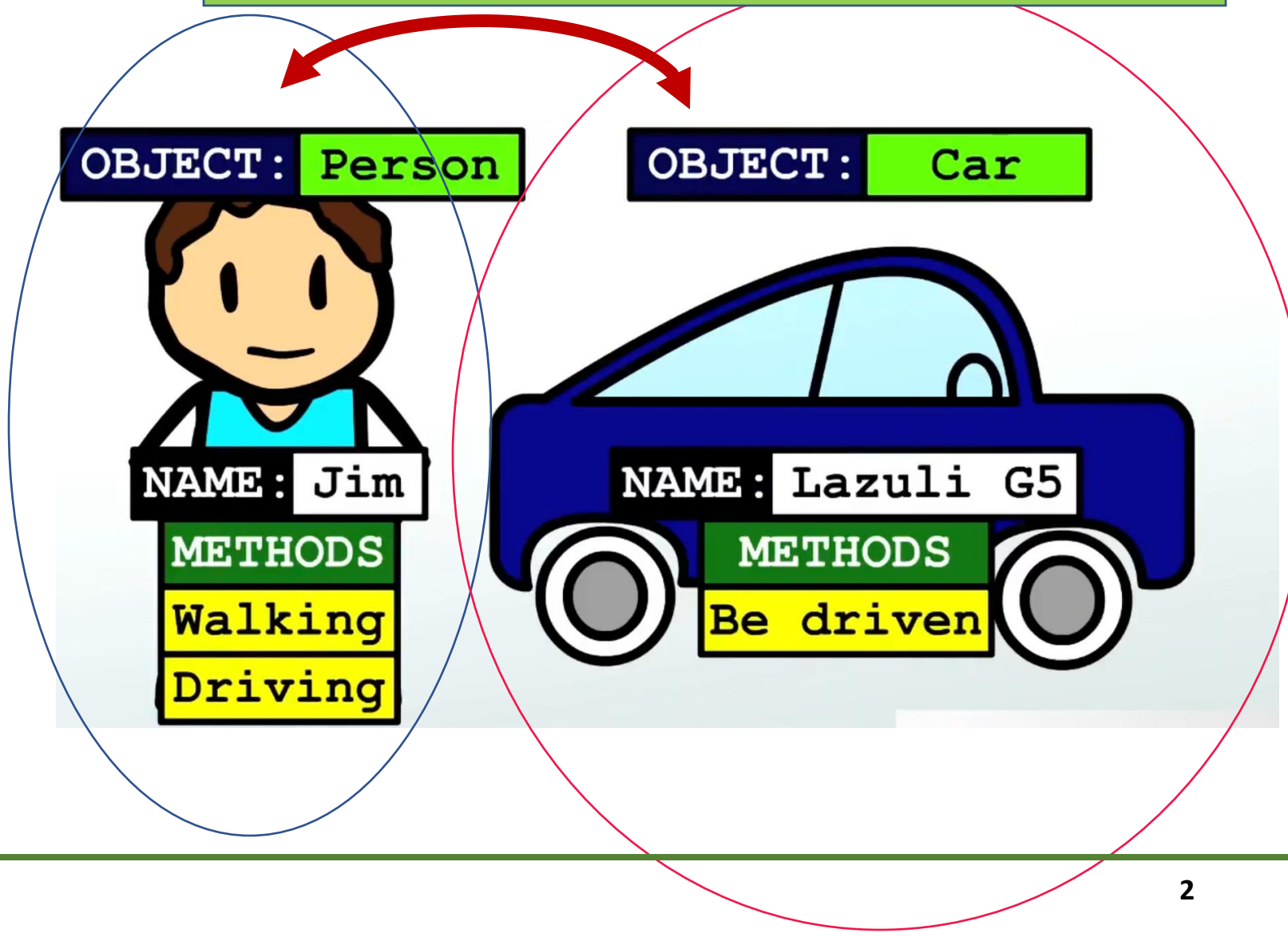
Object Oriented Paradigm



Topic: OOP Concepts

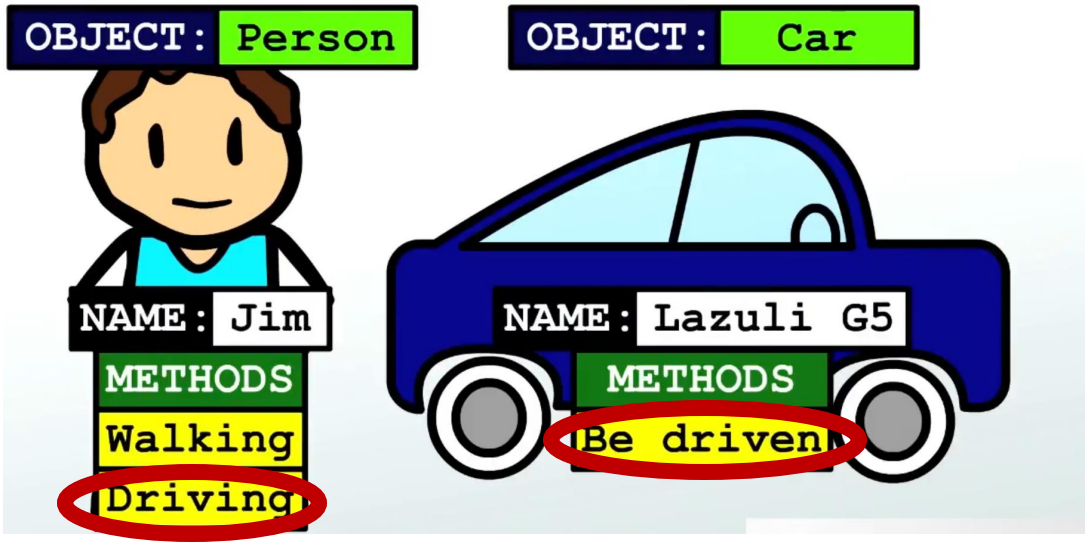
Object

Objects interact with each other to have something happen.



A program where a person get into a car and drive from A to B.

- Includes 2 things:
  - Object 1: Person and
  - Object 2: Car
- Person knows how to drive a car
- And a car knows where its going to be driven.

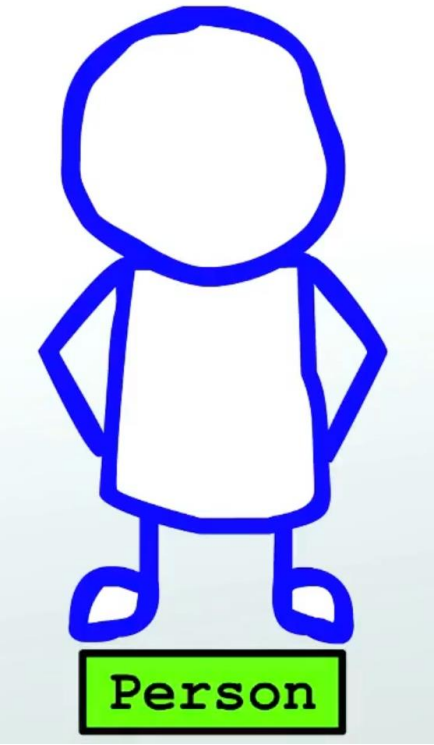


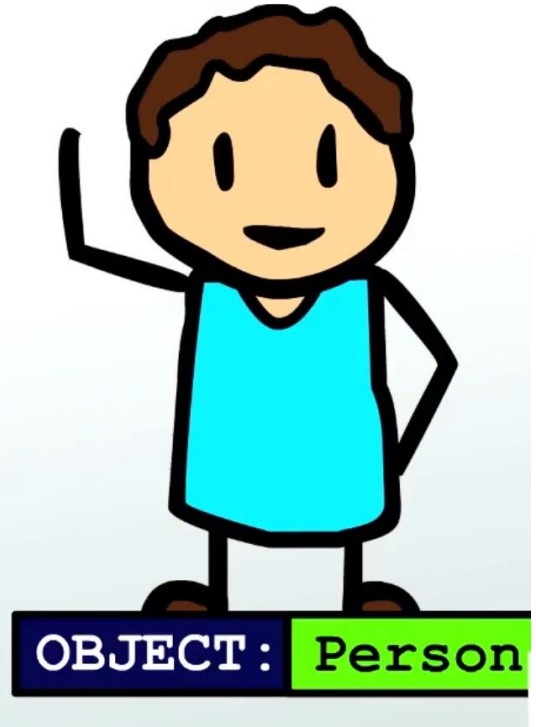
Class

You should have class before you create an Object.

Blueprint of an object

To use class person in a program, you should create an object.

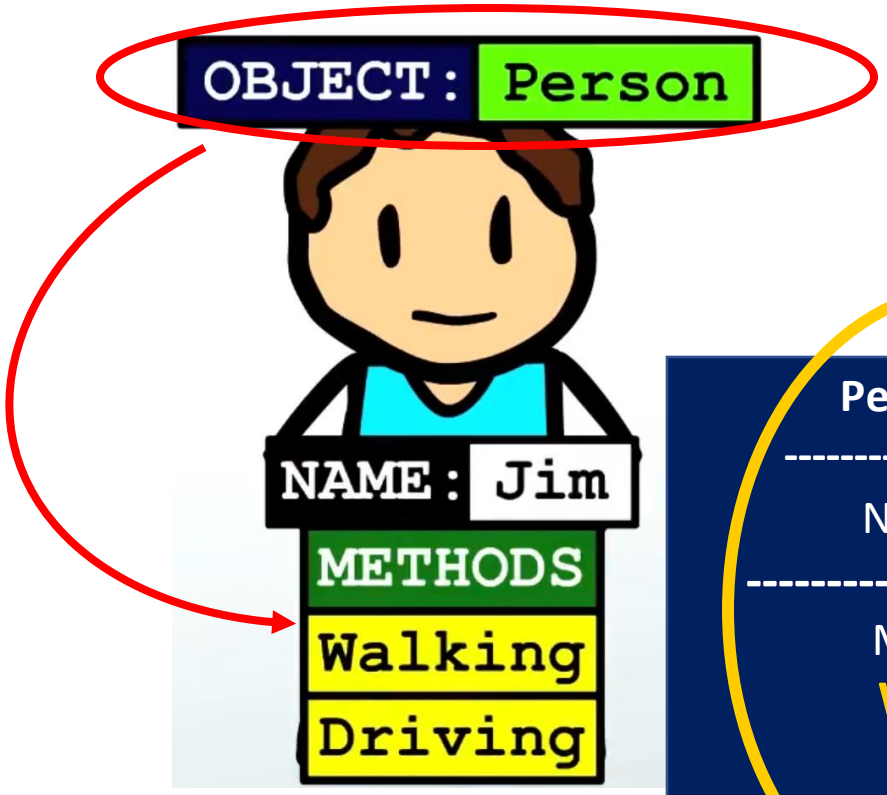




Method

Procedure associated with a class

To make OBJECT do something



Class Person

Person 1:

Name: **Jim**

Methods:
Walking
Driving

Person 2:

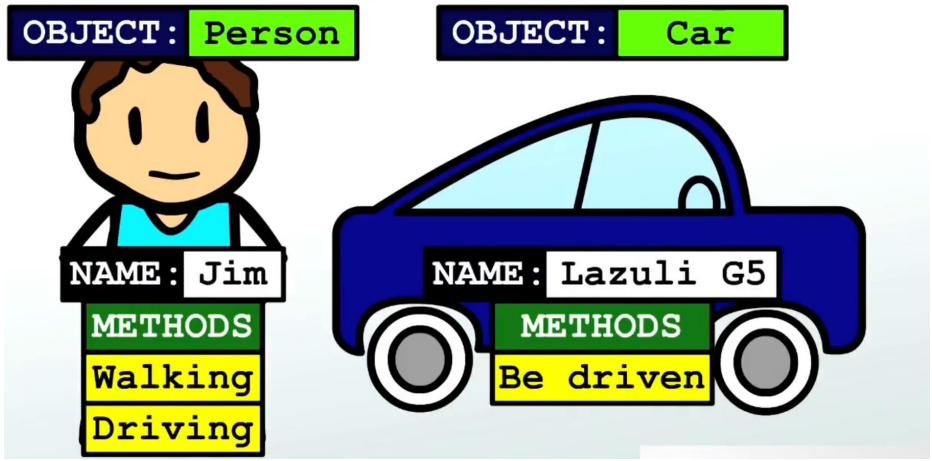
Name: **Mary**

Methods:
Walking
Driving

Objects and Classes

- **Classes** are blueprint or a set of instructions to build a specific type of object.
- Class in Java determines how an object will behave and what an object will contain.

- **OBJECT** is an instance of a class.

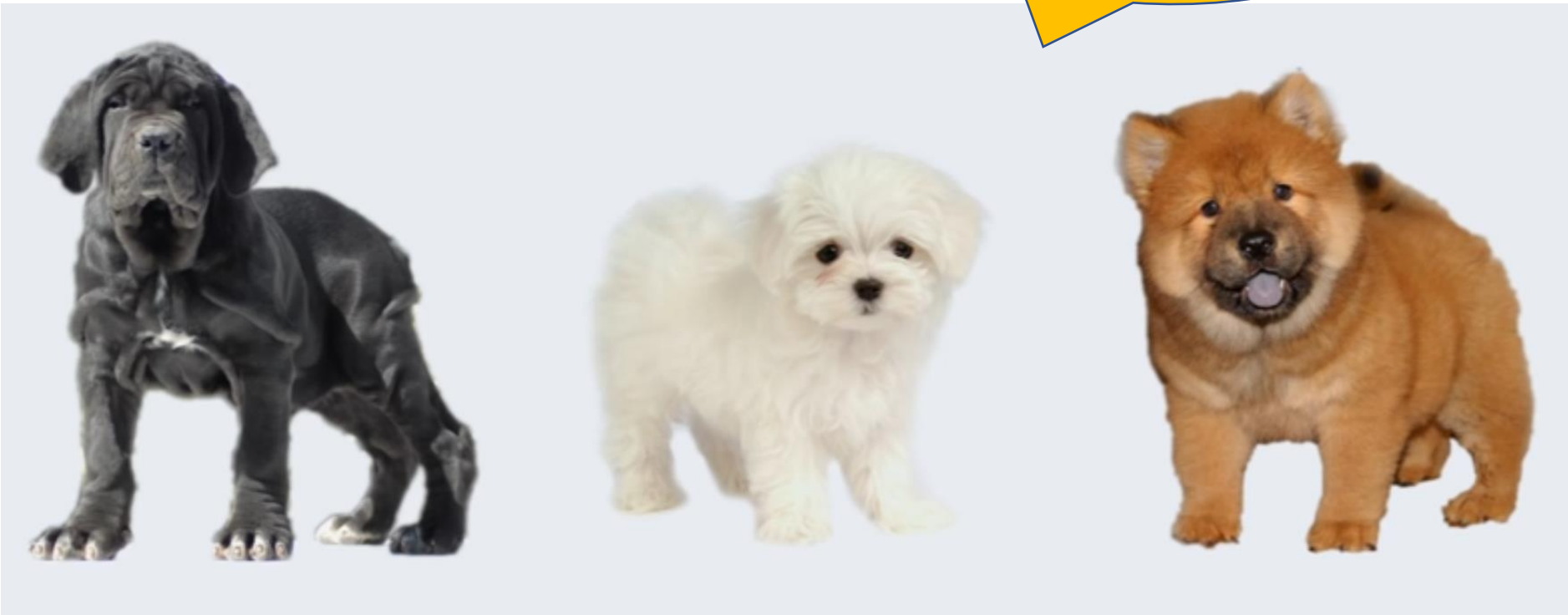


Dog

Common Characteristics shared by dogs:

Data Members
Identified

- ❑ Breed
- ❑ Size
- ❑ Age
- ❑ Color

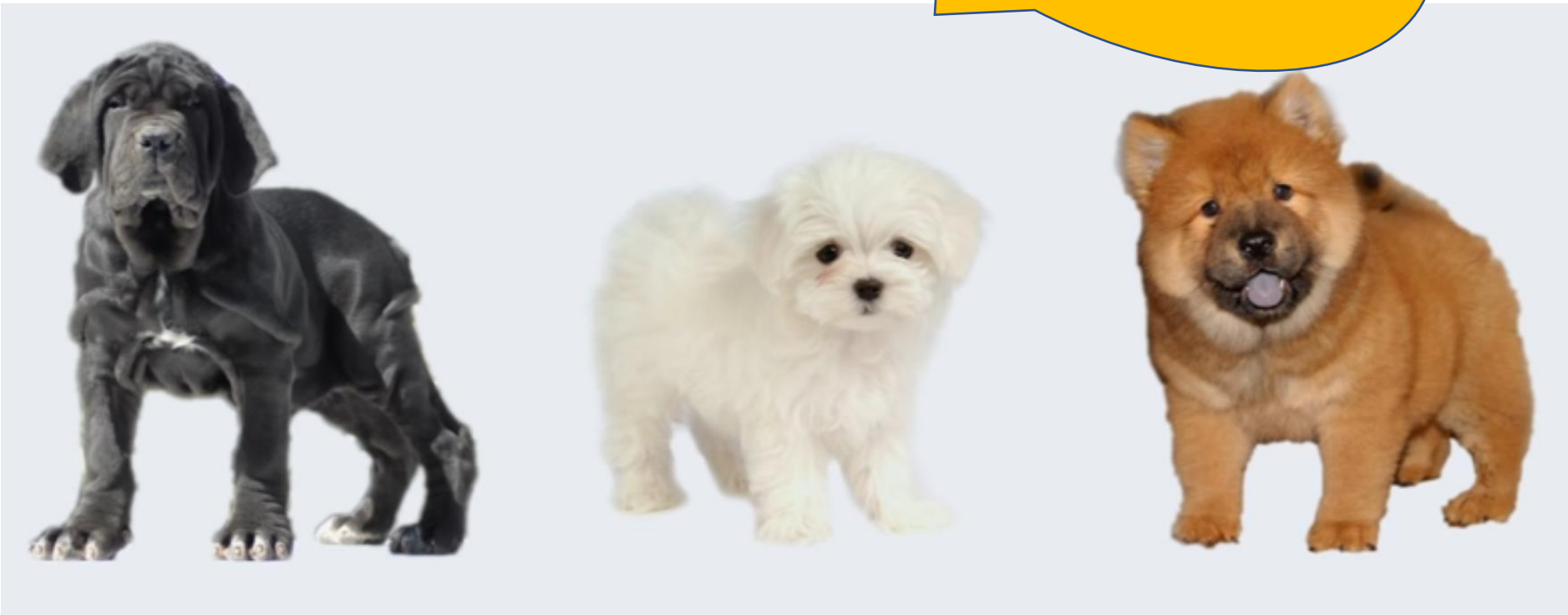


Dog

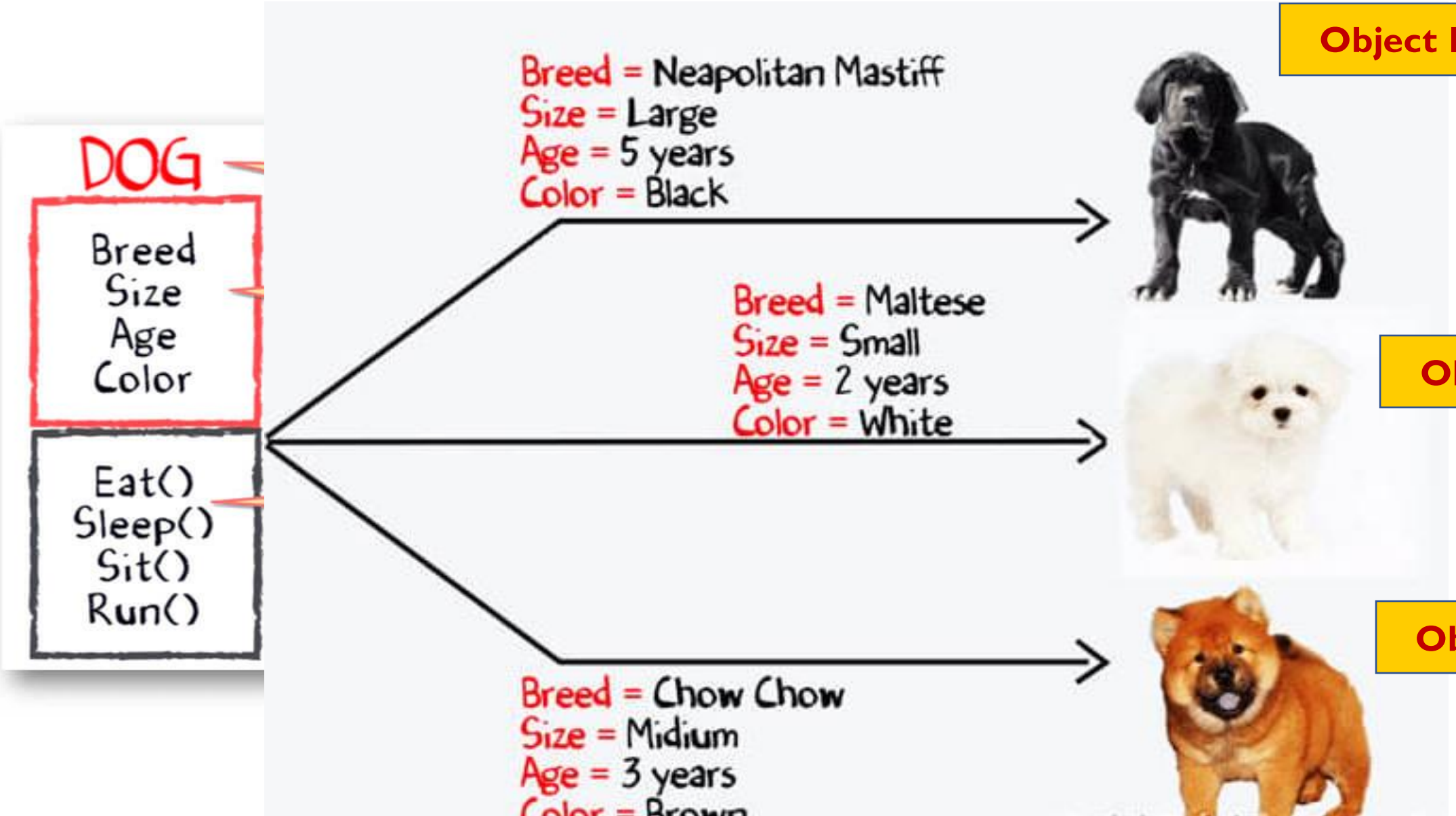
Identity Common Behaviours

Actions
Identified

- ❑ Eat
- ❑ Sit
- ❑ Sleep
- ❑ Run



Classes, Objects and Methods



OOP Concepts

- ❑ Abstraction
- ❑ Encapsulation
- ❑ Inheritance
- ❑ Polymorphism.

Abstraction

Top 10 information of your customer to create a banking application.

- ❑Full name
- ❑Address
- ❑Contact Number
- ❑Tax Information
- ❑Favourite Food
- ❑Favourite Movie
- ❑Favourite Actor
- ❑Favourite Band

Amount of Information available as an individual is infinite.

We might not need all these customer information for a banking application.

Abstraction

We are selecting data from a larger pool.

- ❑ Full name
- ❑ Address
- ❑ Contact Number
- ❑ Tax Information

Same data once abstracted can be used in wide applications

- ❑ Banking
- ❑ Employee
- ❑ Hospital
- ❑ Government

Abstraction

- Only the essential details are displayed to the user.
- The non-essentials units are not displayed to the user.

Registration

- Vehicle Identification Number
- License plate
- Current Owner
- Tax due, date

Owner

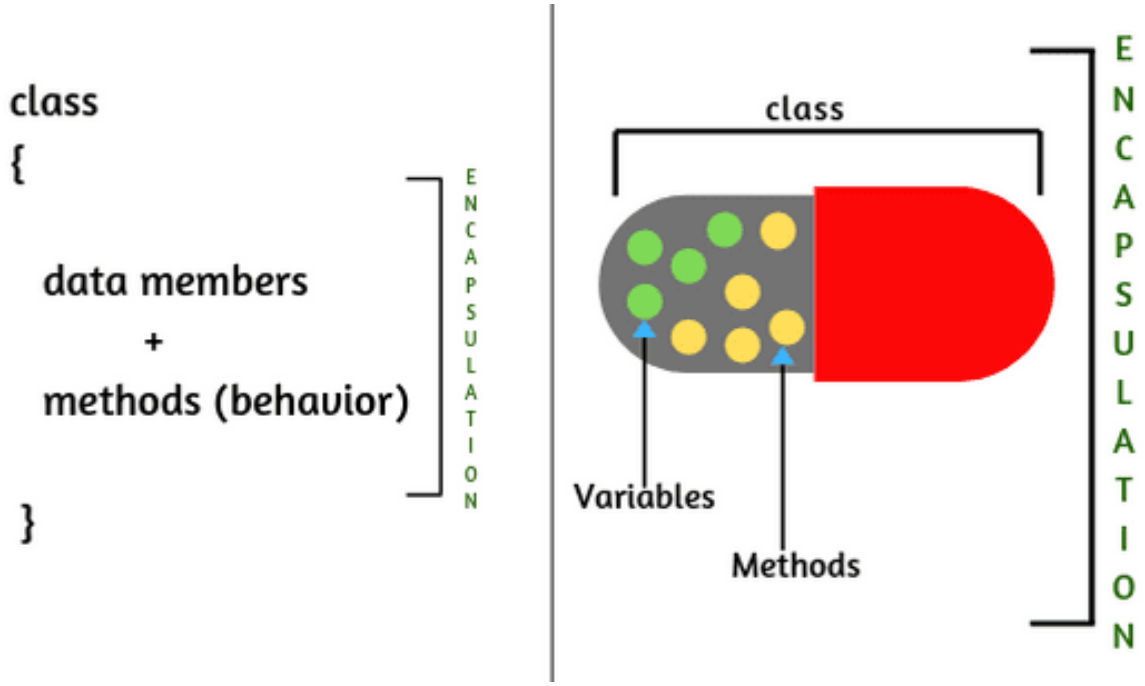
- Car Description
- Service History
- Petrol Mileage History

Garage

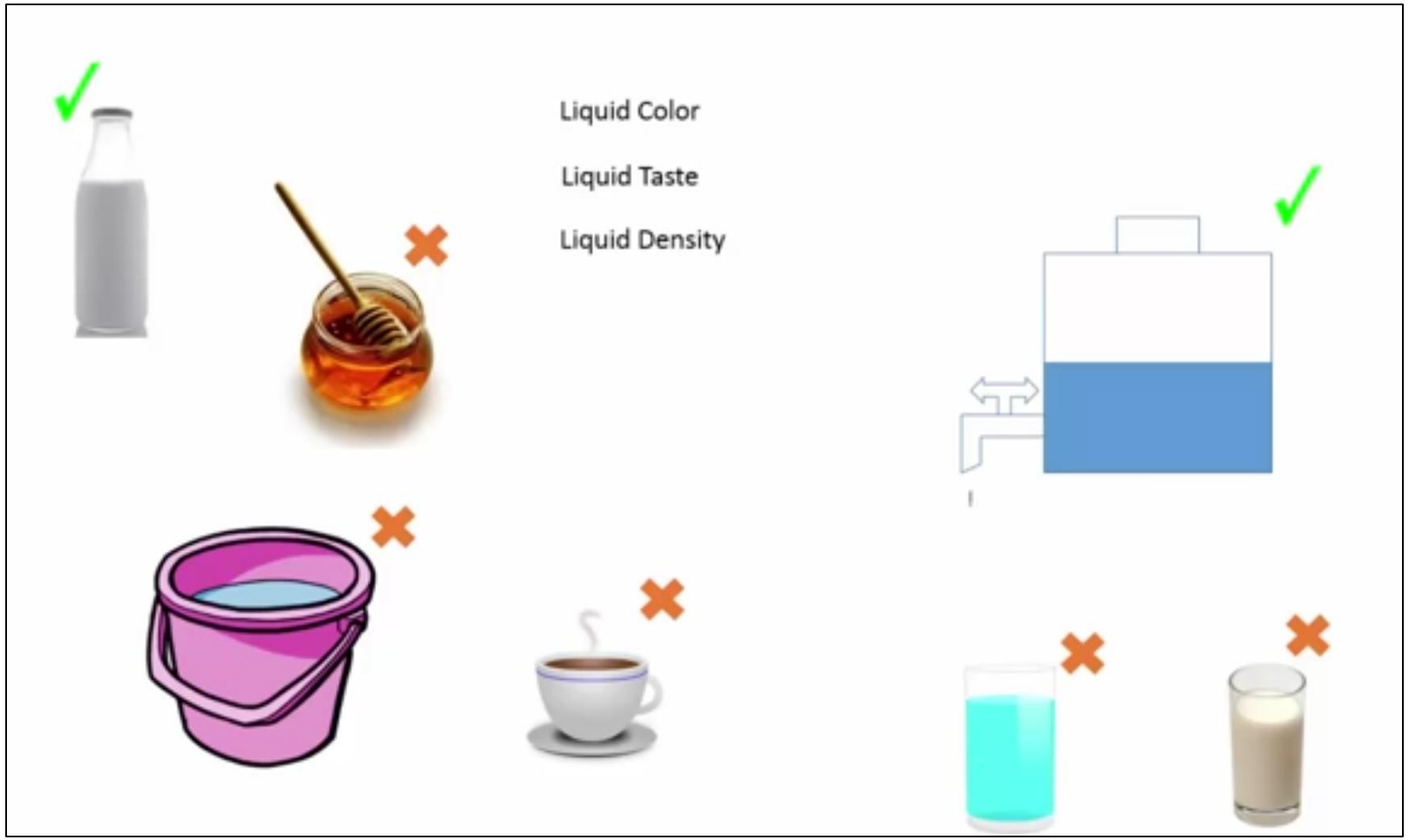
- License plate
- Work Description
- Billing Info
- Owner

Encapsulation

- A mechanism to wrap up variables(data) and methods(code) together as a single unit.
- It is the process of hiding information details and protecting data and behavior of the object.



Why??



Because there is restricted access to liquid inside.

Medicines

Pills

Medicines are covered inside capsule to protect it from any contamination.

Encapsulation

Pill has (encapsulated) **medicine**

Entity

Property

What

How

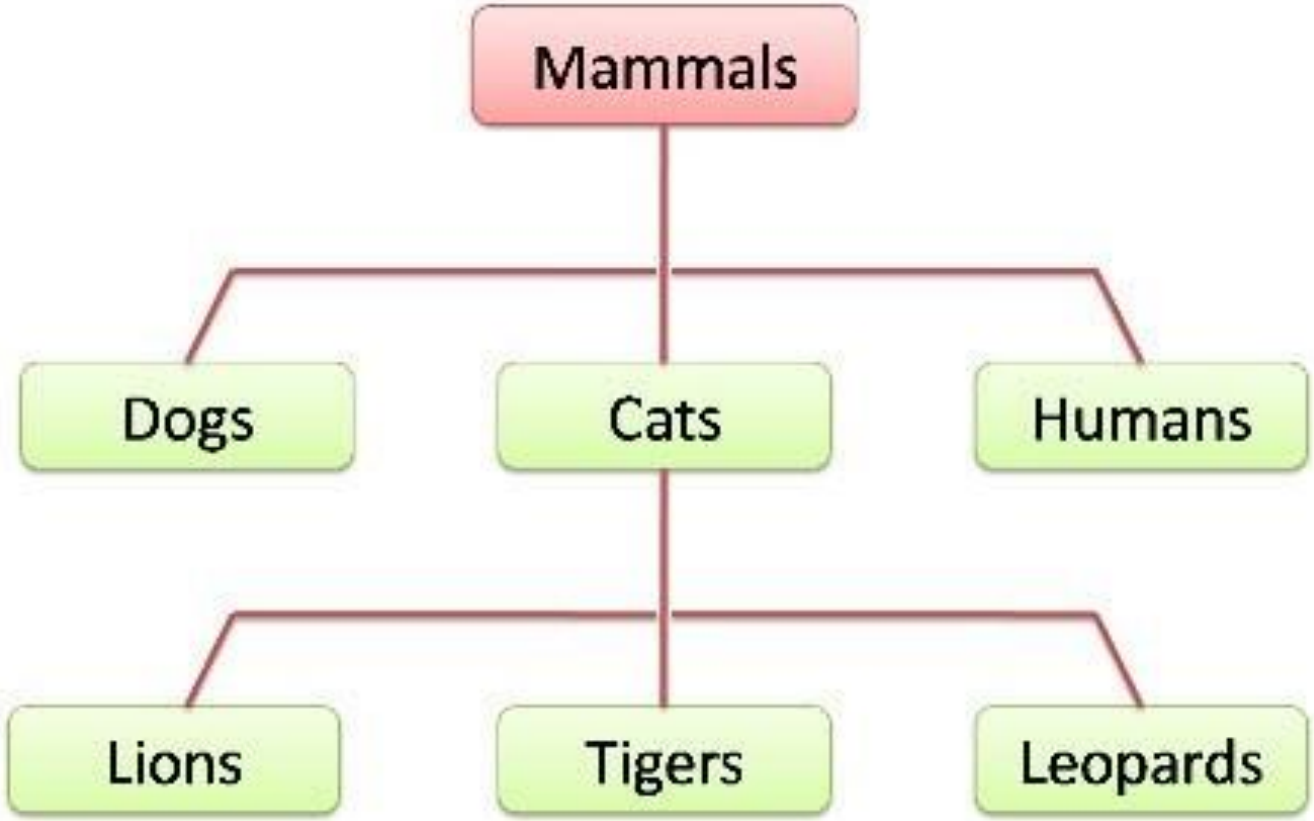
How is **abstraction** and **encapsulation** related??

Identifying entities, properties and behaviours

Process of providing physical structure and mechanism to implement outcome of abstraction.

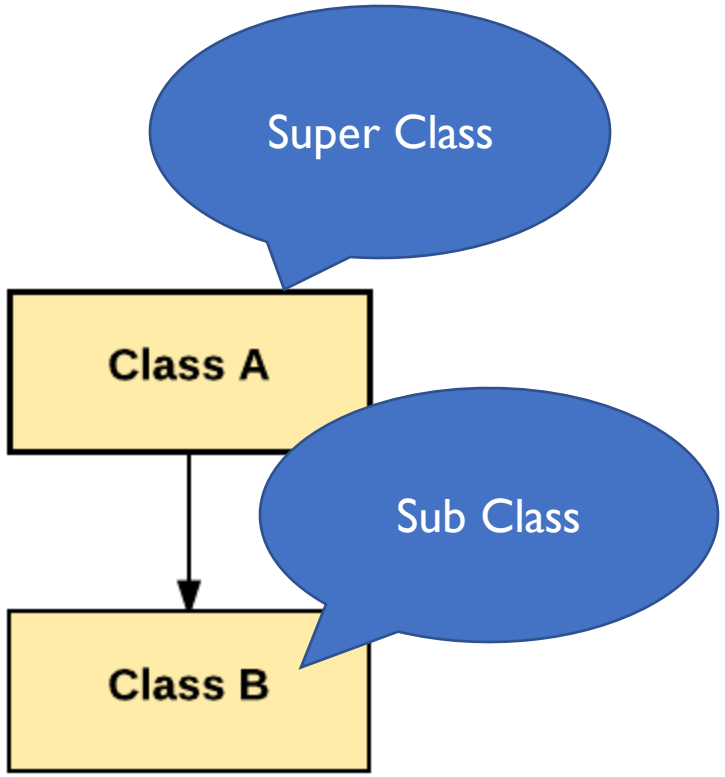
Inheritance

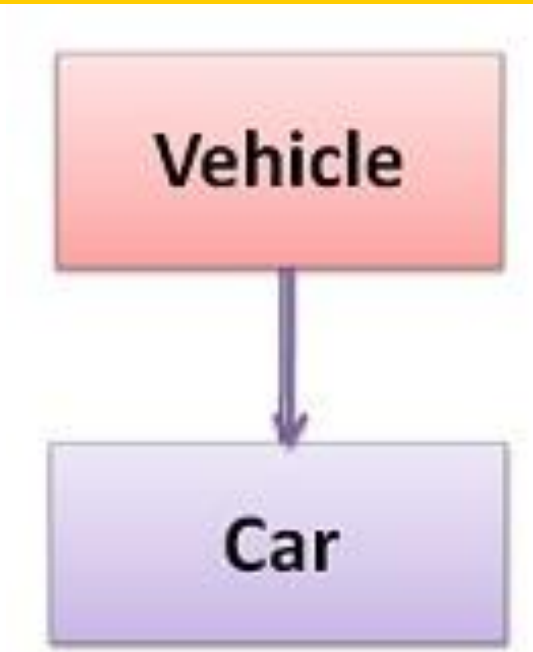
A mechanism in which one class acquires the property of another class.



Single Inheritance

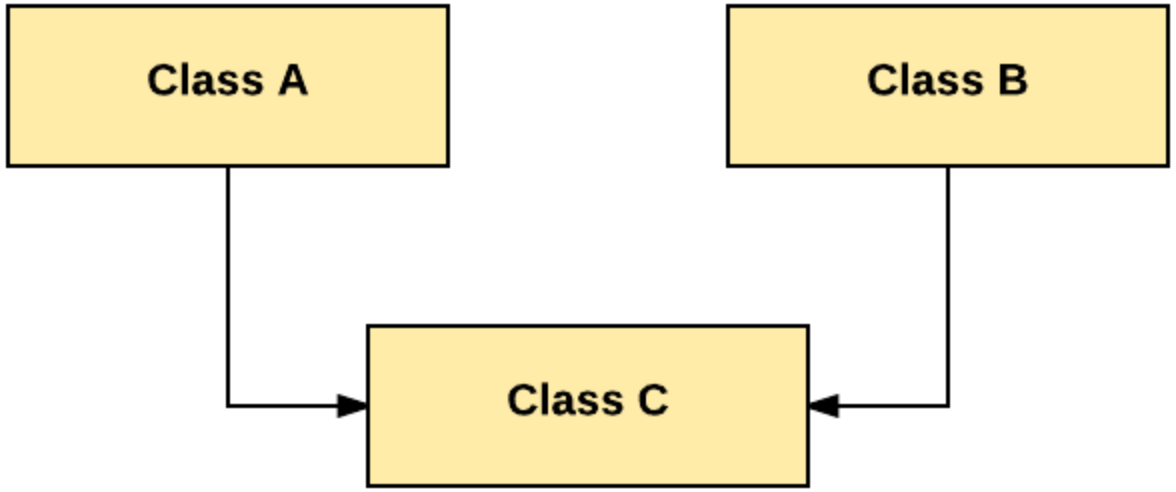
- One class extends another class
- Class B extends only Class A.

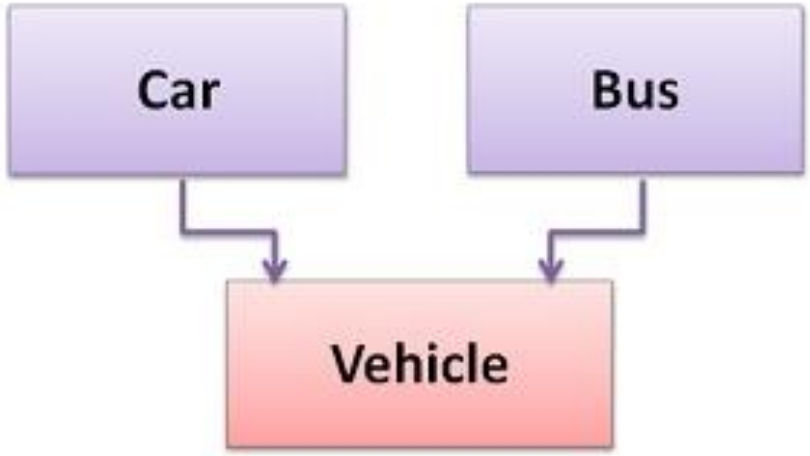




Multiple Inheritance

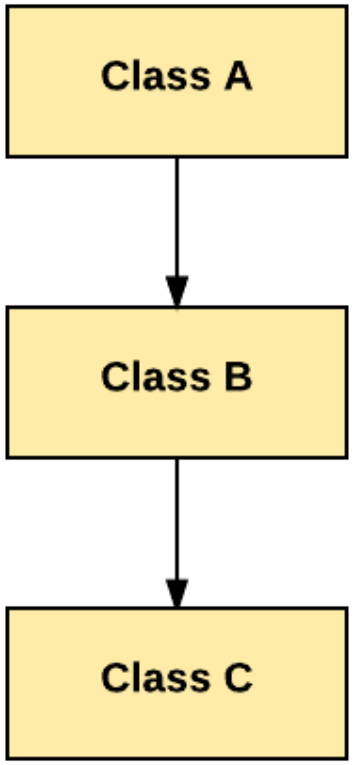
- One class extending more than one class. Java does not support multiple inheritance.
- Class C extends Class A and Class B both.

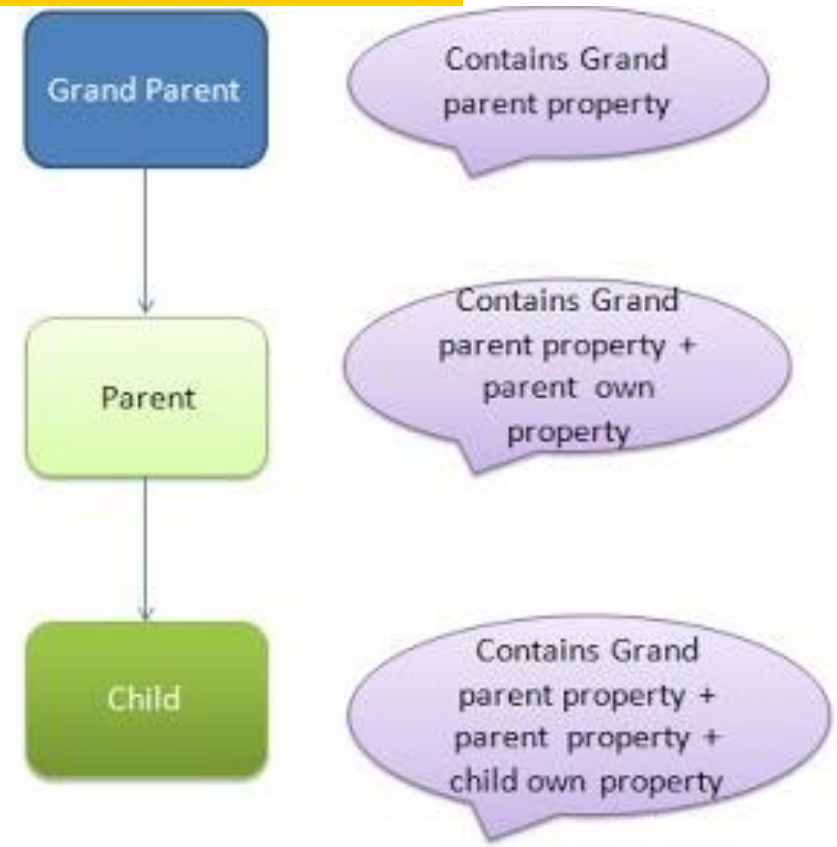




Multilevel Inheritance

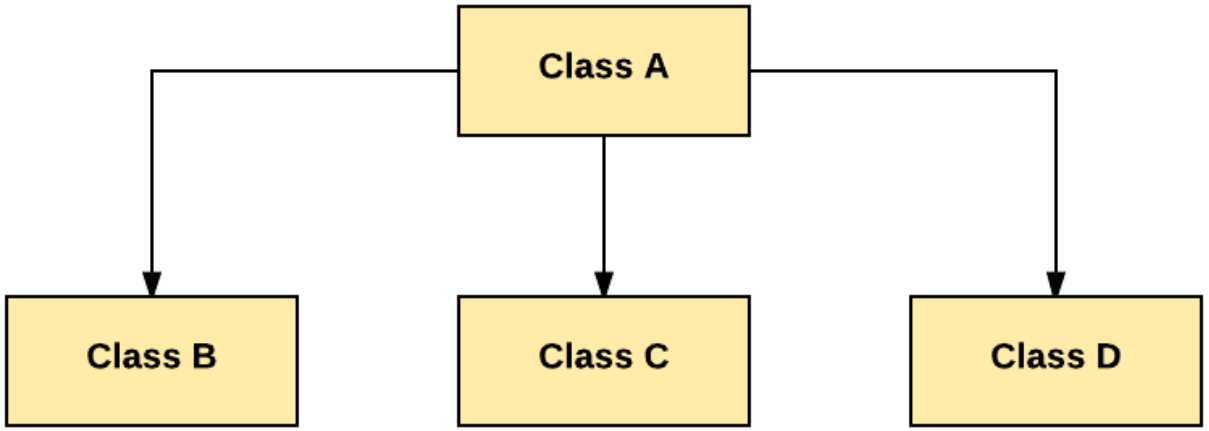
- One class can inherit from a derived class.
- Class C is subclass of B and B is a subclass of Class A.

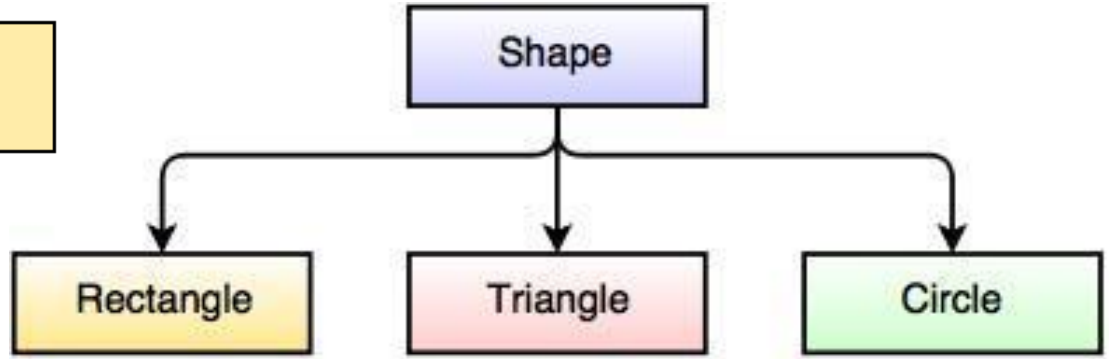




Hierarchical Inheritance

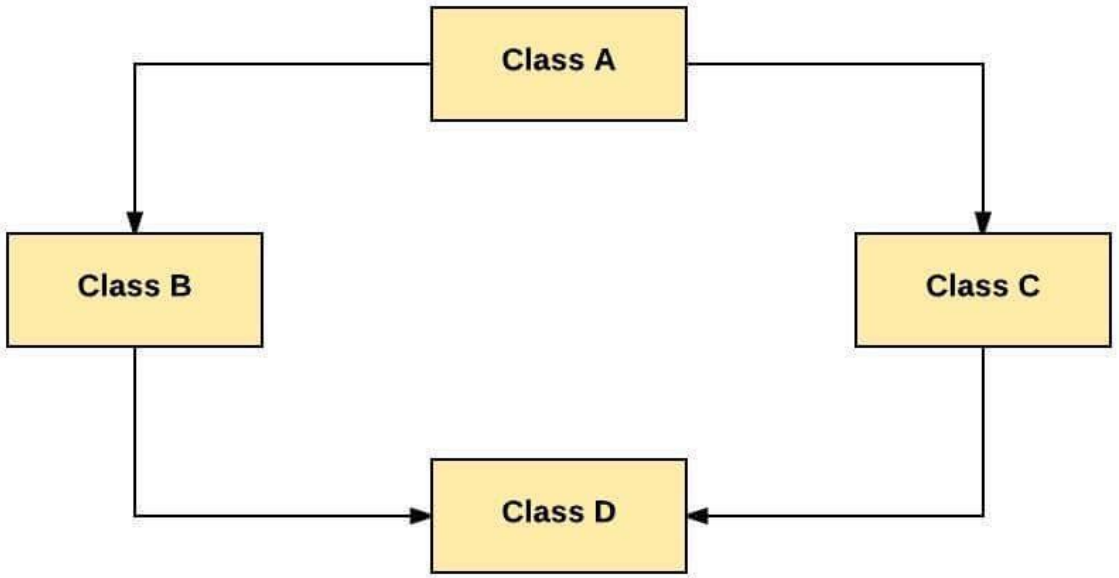
- One class is inherited by many sub classes.
- Class B, C, and D inherit the same class A.

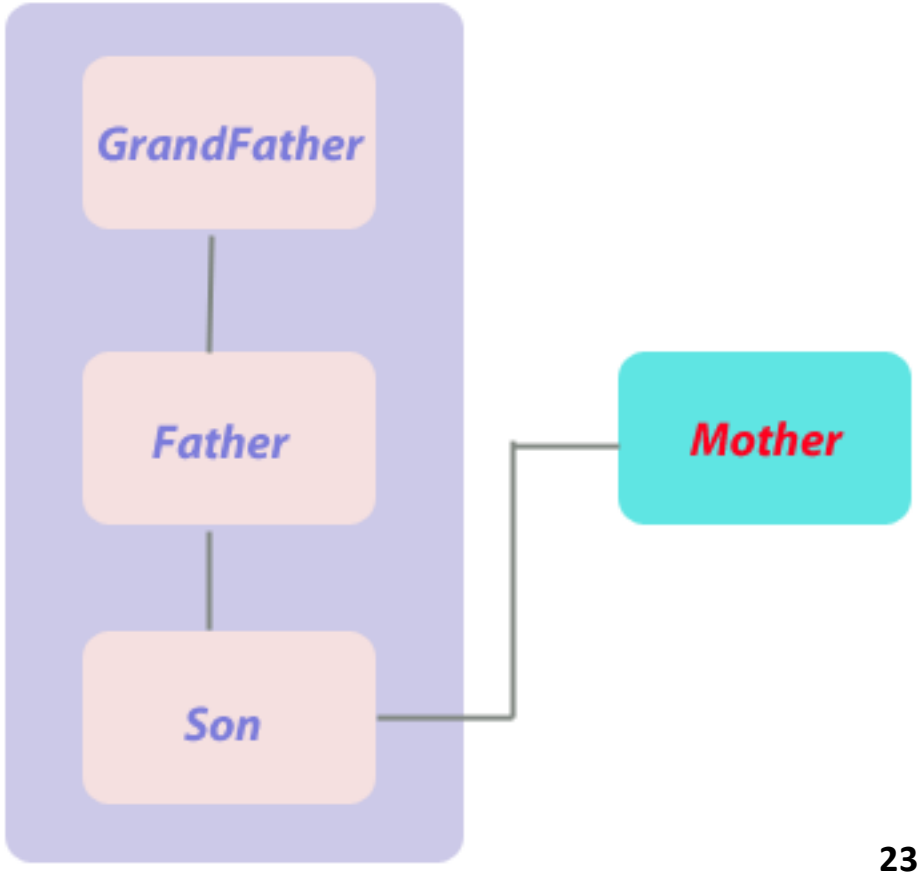




Hybrid Inheritance

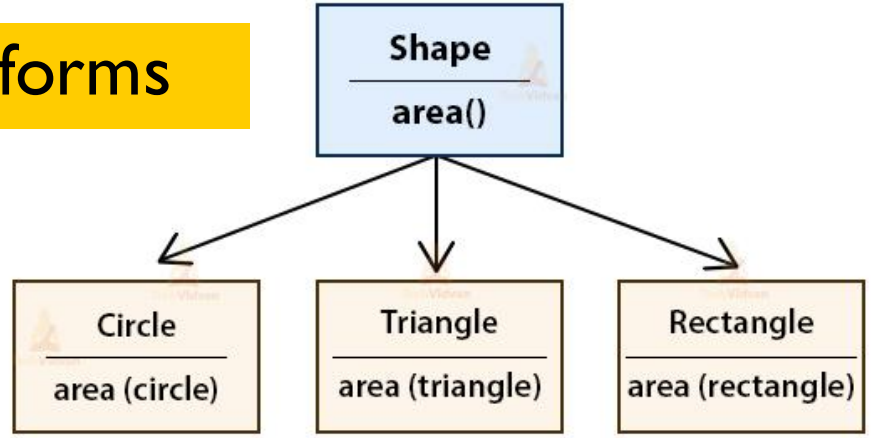
- Hybrid inheritance is a combination of other inheritances.

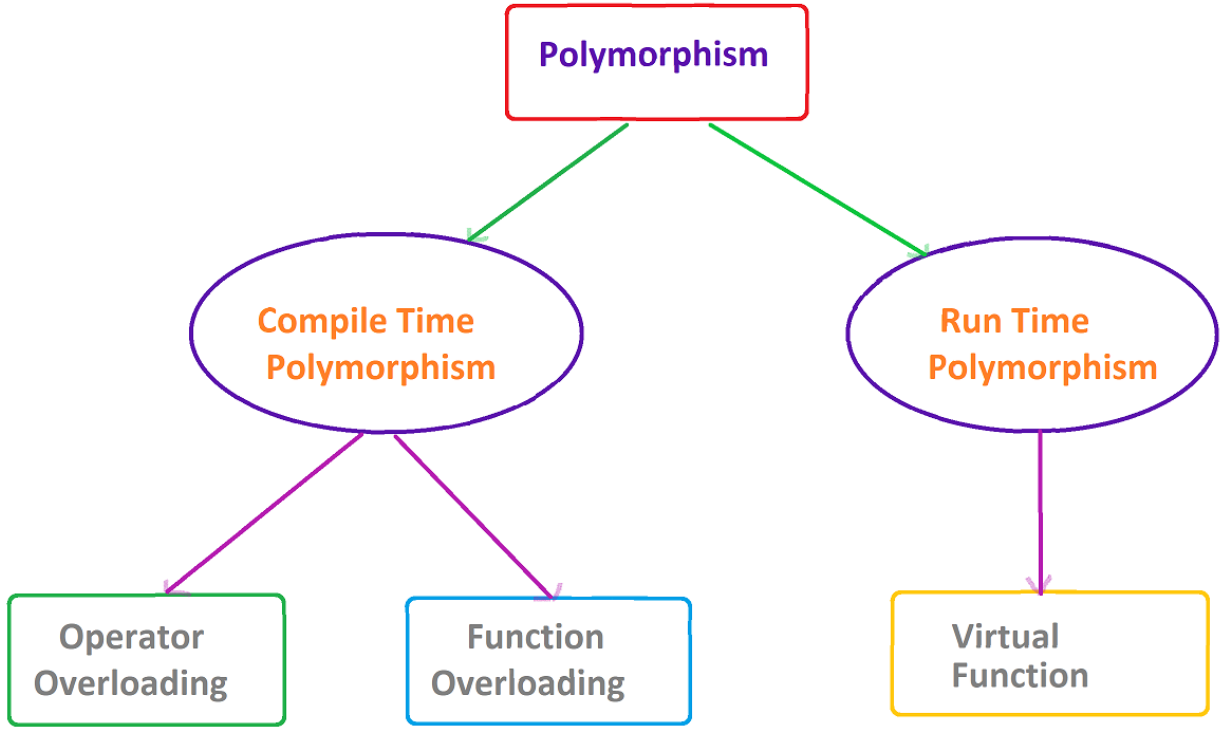


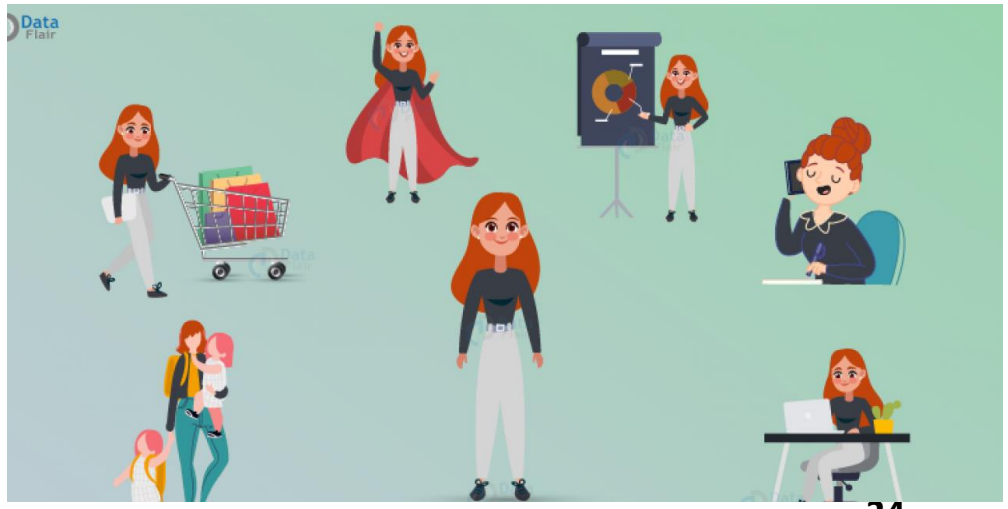


Polymorphism

- The ability of an object to take on many forms







Unit 1

Structured to Object Oriented Approach by Examples, Object Oriented languages, Properties of Object Oriented system, UML and Object Oriented Software Development, Use case diagrams as a functional model, Identifying Objects and Representation by Object Diagram: state and behaviour, Identifying classes and CRC Cards, Simple Class using class diagram: Encapsulation, Data Hiding, Reading and Writing Objects, Class Level and Instance Level Attributes and Methods, Generalization using Class Diagram: Inheritance, Constructor and Over Riding, Visibility: Attribute, Parameter, Package, Local and Global.